When does spatial smoothing help anomaly scoring on lattices of embeddings? Fixed windows, similarity-weighted aggregation and order gates

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Abstract

Anomaly scores on lattices of embeddings, such as the patches of an image, are often smoothed over a fixed spatial window. On lattices simulated with MyUncle, on synthetic images and on a controlled generator that varies only the size and shape of the anomalies, a fixed 3×3 mean destroyed the signal for scattered and bar-shaped anomalies and helped only with DINOv2 embeddings. Aggregation weighted by the cosine similarity of the embeddings of neighboring sites, with a label-free scale, preserved scattered, bar-shaped, irregular and boundary anomalies: on independent test scenarios it raised the lower-tail average precision from 0.11 (fixed mean) and 0.23 (no smoothing) to 0.32, with paired 95% intervals above zero. Gates based on local order and on the von Mises–Fisher information of the neighborhood, a d-dimensional extension of MyUncle's entropy quantities, did not beat ordinary similarity; rank gates on that information coincide with gates on the resultant length. A contrast-survival model, in which a fixed window keeps only the interior of an anomaly, predicted the loss (Spearman 0.82). A GaaR search over 460 rule configurations did not beat random search. All evidence is simulated; code and results are open.

Keywords: anomaly detection, spatial aggregation, embedding lattices, von Mises–Fisher distribution, orientational order, configuration search

1. Introduction

Many pattern recognition problems put a vector on every site of a grid: the embeddings of the patches of an image, or the orientation field of an ordered medium. A common way to turn per-site anomaly scores into regions is to smooth them over a fixed window, which suppresses isolated false alarms. We ask when that helps. Unsupervised anomaly detection has been developed for large surveys in astronomy (Baron and Poznanski, 2017; Lochner and Bassett, 2021) and in medical imaging, mostly as detection of deviation from a healthy appearance (Schlegl et al., 2019); spatial aggregation of the scores is a routine post-processing step, and its failure modes are rarely measured.

MyUncle (Sandler, 2026a) is a simulation framework for a maintenance lattice (rotors whose order is eroded by a state-dependent flux and restored by repair) with an order-entropy bridge. This paper uses its lattices as a testbed and its order quantities as a candidate gate. Contributions: (1) a controlled failure analysis, on lattices simulated with MyUncle, on synthetic images and on an independent generator that varies only the size and shape of the anomalies, showing that a fixed window can destroy the signal; (2) aggregation weighted by the similarity of the embeddings of neighboring sites, with a label-free scale, which preserved scattered, bar-shaped, irregular and boundary anomalies on independent test scenarios, with paired confidence intervals; (3) a negative result for gates based on local order and on the von Mises–Fisher information of the neighborhood (the d-dimensional extension of MyUncle's entropy quantities, verified against the unmodified code), which did not beat ordinary similarity; (4) a comparison of GaaR (Sandler, 2026c), an evolutionary search framework, with random search for selecting among rule configurations; and (5) an open pipeline (Sandler, 2026b). All evidence is simulated; no experimental image was analyzed.

2. Scores and aggregation rules

2.1. Rarity score

Each site i carries a unit vector u_i. For images, a patch is a 224×224 px window (0.5 μm/px) encoded by DINOv2 ViT-S/14 (Oquab et al., 2023) (384-d, pretrained on natural images) or by a deterministic handcrafted encoder of color, gradient-orientation and texture statistics (64-d; Ruifrok and Johnston, 2001); embeddings are centered, reduced by a cohort PCA (16 components; Pedregosa et al., 2011) and normalized. For simulated lattices the vector is the rotor $(\cos\theta, \sin\theta)$, optionally with the local disorder as a third coordinate; the intact flag is never part of it. The rarity r_i is the cohort percentile of the mean cosine distance to the k nearest sites of a reference sample (same-image neighbors within six steps excluded), evaluated with k equal to 0.5, 1, 2, 5 and 10% of the sample, the highest rank counting, because the prevalence of an interesting state is not known in advance.

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2.2. Aggregation rules

A rule turns r into the final score s using the lattice neighbors N_i in a $w \times w$ window and the unit embeddings, never labels. The frozen configuration of earlier versions of this work is the fixed mean, $s_i = |W_i|^{-1} \sum_{j \in W_i} r_j$ with $w = 3$. Similarity-weighted aggregation weights each neighbor by the compatibility of its embedding with that of site i,

$$c_{ij} = \exp(-(1 - u_i^\top u_j)/\tau), \qquad s_i = \frac{\sum_{j \in W_i} c_{ij}\, r_j}{\sum_{j \in W_i} c_{ij}}, \tag{1}$$

with τ the median of $1 - u_i^\top u_j$ over lattice-adjacent pairs of the cohort (label-free, no free parameter). It is an edge-preserving filter in embedding space of the bilateral kind; we make no claim to its novelty. Gated variants mix the unsmoothed and the aggregated score, $s_i = (1 - g_i) r_i + g_i\, s_i^{\mathrm{sim}}$ with $g_i = \sigma(\alpha(A_i - \vartheta))$, where A_i is the cohort rank of a local statistic of the neighbors of i: the mean compatibility c_{ij}, the resultant length R_i (one minus MyUncle's local disorder) or the von Mises–Fisher order information H_χ. A one-sided rule, $s_i = r_i + \max(0, m_i - r_i)$ with m_i the similarity-weighted mean over the neighbors, lets neighbors add evidence but never remove it; it is the asymmetric rule one uses for a failure mode that has a sign. A nucleation-and-growth rule with hysteresis lets a site whose rarity exceeds a nucleation threshold (a cohort rank) seed, and raises weaker neighbors to the seed's score weighted by their compatibility, never lowering a score; it combines the two-threshold logic of hysteresis thresholding (Canny, 1986) with embedding compatibility as the growth probability. An optional persistence factor multiplies s by $(1 - \lambda) + \lambda P_i$, where P_i is the share of the windows 1, 3, 5, 7 at which the score exceeds its cohort top-3% threshold.

2.3. Order and entropy

MyUncle's local disorder is $D_i = 1 - R_i$ with $R_i = \left| \frac{1}{|N_i|} \sum_{j \in N_i} u_j \right|$, valid in any dimension. The concentration κ_i of the von Mises–Fisher distribution on S^{d-1} (Banerjee et al., 2005; Mardia and Jupp, 2000) with mean resultant length R_i solves $A_d(\kappa) = I_{d/2}(\kappa)/I_{d/2-1}(\kappa) = R_i$, and the information held in the local order is $H_\chi = S_{\mathrm{unif}} - S_{\mathrm{vMF}}(\kappa, d)$ with

$$S_{\mathrm{vMF}} = -(\tfrac{d}{2} - 1) \ln \kappa + \tfrac{d}{2} \ln(2\pi) + \ln I_{d/2-1}(\kappa) - \kappa A_d(\kappa). \tag{2}$$

For $d = 2$ these are MyUncle's quantities; the tests agree with the unmodified core to 10^{-12} (disorder) and 10^{-9} (entropy; Fig. 1). These are a mathematical characterization of local order. H_χ is a monotone function of R_i, so a gate on the *rank* of H_χ is identical to a gate on the rank of R_i; the detector's behavior below comes from the rarity score and the aggregation rule, not from the entropy.

3. Evaluation design

Scenarios. (i) Synthetic stained-tissue images (4480×4480 px; tubule-like rings, glomerulus-like discs, vessels, stain variation, planted anomalies and artifacts; 8 development and 16 held-out images, the latter with three anomaly types not used in development), each with the handcrafted and the DINOv2 embedding. (ii) Lattices simulated with MyUncle's kinetic Monte Carlo (unmodified core; 48×48 sites, snapshots at steps 20, 40, 60, damaged sites as ground truth), with the rotor and rotor + D embeddings; the development preset is chiral scaffold, the test presets are superconducting scaffold, polymer photodegradation, liquid crystal and magnetic storage decay, which are model parameter sets, not fitted to measurements. (iii) A controlled generator of embedding lattices (32×32 sites, 16 dimensions, smooth background domains, 3% anomalous sites of fixed separability, 10 images per cohort, 5 cohorts) in which only the anomaly changes: scattered single sites, 3×3 and 6×6 blocks, bars of 8 sites, irregular 12-site clusters, and 3×3 blocks on domain boundaries; each anomaly has its own direction; the separability was calibrated beforehand on the unsmoothed score only. Development and test scenarios use disjoint seeds, and the test images and lattices are new.

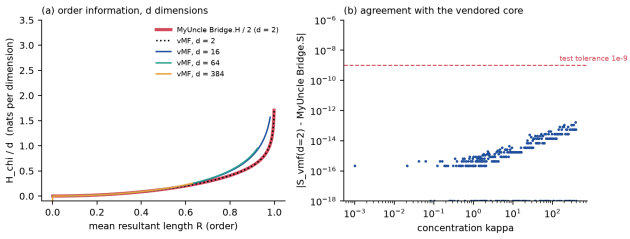


Figure 1: (a) The order-information bridge in d dimensions: MyUncle's H_χ ($d = 2$, red) and the von Mises–Fisher generalization for $d = 2$, 16, 64 and 384. (b) Difference between the $d = 2$ entropy and the vendored MyUncle function.

Protocol. The rarity score is computed once per scenario (the same for every rule). Rule parameters (rule, window 3/5/7, ϑ, α, λ; 460 distinct configurations) were selected on the ten development scenarios by $F = \frac{1}{2}\overline{\mathrm{AP}} + \frac{1}{2} \min \mathrm{AP}$ (average precision against the ground truth, lower-tail across scenarios). The table of all configurations was computed exhaustively, so a search can be compared exactly with the optimum: each GaaR strategy and plain random search ran 100 times with a budget of 40 evaluations. The champions and an untuned similarity-weighted rule were then evaluated once on the ten independent test scenarios; 95% intervals are paired bootstrap intervals that resample the independent units (images; simulation runs of three snapshots) with the same draws for every rule. Selecting rule parameters uses development labels; the rules themselves are label-free.

4. Results

4.1. A fixed window can destroy the signal

On the independent test scenarios (Table 3) the fixed 3×3 mean lowered the average precision of scattered single-site anomalies from 0.37 to 0.03, of 8-site bars from 0.39 to 0.05 and of the simulated lattices (rotor embedding) from 0.56 to 0.32; the region recall of scattered anomalies at the top 3% fell from 0.38 to 0.03 and that of bars from 0.58 to 0.07. The loss shrank as the anomalies became compact, and the mean helped only the DINOv2 embeddings of the synthetic images (0.11 to

0.50), whose unsmoothed score is noisy. Averaged in the lower-tail fitness F, the fixed mean (0.11) was below no smoothing (0.23) at the test (paired difference −0.12 [−0.19, −0.07]).

4.2. Contrast survival

A simple model explains the failure. Let anomalous sites have the mean rarity rank r_a and normal sites r_n (calibrated on the development cohorts: 0.98 and 0.65; the ranks saturate near 1 because the highest of five scales counts). After the $w \times w$ mean, a site with k anomalous among its n window sites scores $(kr_a + (n - k)r_n)/n$, and it stays in the top 3% of the cohort, whose threshold θ is read from the smoothed scores themselves (label-free, 0.97 on average), only if $k \geq k^* = n(\theta - r_n)/(r_a - r_n)$; here $k^* \approx 8.7$ of 9 sites, so practically the whole window must be anomalous. The fixed mean therefore preserves only the interior of an anomaly, the sites at least $(w - 1)/2$ steps from its border, and erases anomalies thinner than the window; for a square block of side b the surviving share is $((b - w + 1)/b)^2$, and for scattered sites and bars it is zero. The predicted share of surviving anomalous sites, computed from the true masks, correlated with the measured average precision of the fixed mean over the 60 cohorts of the controlled generator (Spearman ρ = 0.82; test cohorts only 0.76; 5×5 window 0.64) and reproduced the order of the shapes (Table 1). The model is explanatory: r_a and r_n were calibrated with labels on the development cohorts, and it describes the fixed mean only. It also explains why similarity weighting works: neighbors with dissimilar embeddings receive small weights, so a site at the border of an anomaly is averaged mainly with anomalous neighbors.

A label-free version replaces the masks by the score field itself: the candidates are the sites of the top 3% of the unsmoothed rarity, and a candidate survives when all sites of its 3×3 window lie in the top 9%, so that the predictor is the interior of the candidate set. It correlated with the measured average precision of the fixed mean over the 60 cohorts (Spearman ρ = 0.56; test cohorts only 0.51; 0.61 and 0.55 when the neighbor threshold is the top 6% or 15%) and with the label-based prediction (ρ = 0.65). It tracks the loss of the fixed mean only: it correlated negatively with the average precision of no smoothing and of similarity weighting (−0.39 and −0.42), so it is not a quality predictor in general. In this generator the fixed mean did not beat no smoothing in any of the 60 cohorts, while similarity weighting beat both in all 60 (mean AP 0.41 without smoothing, 0.11 for the fixed mean, 0.52 for similarity weighting); a choice between the fixed mean and no smoothing based on the predictor, fitted on the development cohorts, therefore reached 0.36 on the test cohorts against 0.38 for no smoothing, and the choice is unnecessary once similarity weighting is used.

4.3. Similarity-weighted aggregation

Weighting neighbors by embedding similarity removed most of that loss (Table 3). The untuned 3×3 rule reached F = 0.32, +0.21 [+0.14, +0.30] above the fixed mean and +0.09 [+0.04, +0.19] above no smoothing (paired 95% intervals). It kept the region recall of scattered anomalies at 0.44 and of bars at 0.60, lowered the false regions per image from 2.16 (no smoothing, averaged over the controlled shapes) to 0.57, and raised the DINOv2 image AP to 0.75. It did not help the handcrafted-feature images (0.17, against 0.18 without smoothing), and on the rotor-only lattices it stayed slightly below no smoothing (0.51 against 0.56). The one-sided rule matched no smoothing (+0.00 [+-0.00, +0.01] in F).

Table 1: Contrast survival (window 3×3): predicted share of anomalous sites that survive the fixed mean and the measured average precision (development and test cohorts).

Anomaly	Predicted survival	AP, fixed mean	AP, no smoothing
Scattered sites	0.00	0.03	0.41
3×3 blocks	0.12	0.11	0.40
6×6 blocks	0.41	0.29	0.41
Bars of 8	0.00	0.05	0.41
Irregular clusters (12)	0.01	0.10	0.45
Boundary blocks 3×3	0.11	0.09	0.38

Table 2: Development: the best configuration of every rule family and its development fitness F (460 distinct configurations in all).

Rule family	F_{dev}	Selected parameters
No smoothing	0.30	-
Fixed mean	0.18	window 3, $\lambda = 0.50$
Similarity-weighted	0.41	window 3, $\lambda = 0.25$
One-sided	0.31	window 3, $\lambda = 0.25$
Order gate, compatibility	0.41	window 3, $\vartheta = 0.2$, $\alpha = 5$, $\lambda = 0.50$
Order gate, R_i	0.39	window 3, $\vartheta = 0.2$, $\alpha = 2$, $\lambda = 0.50$
Order gate, H_χ	0.39	window 3, $\vartheta = 0.2$, $\alpha = 2$, $\lambda = 0.50$

4.4. Order gates and the von Mises–Fisher information

The development-selected order gates did not beat ordinary similarity: the gate on compatibility reached F = 0.30, the gates on the resultant length and on H_χ 0.31, against 0.32 for the untuned similarity-weighted rule (−0.02 [−0.04, +-0.00] and −0.02 [−0.03, +-0.00], paired intervals). The last two gave the same scores to within 0.01 in every scenario, as expected from the monotone relation of H_χ and R_i (the small differences come from the numerical inversion of A_d). The d-dimensional entropy extension is therefore reported as a formal result; it did not improve detection in these experiments.

4.5. Two further ideas

The nucleation-and-growth rule, with its parameters selected on the development scenarios, reached F = 0.23 on the test scenarios: the same as no smoothing (paired difference +0.00 [+-0.00, +0.00]) and below the similarity-weighted rule (−0.09 [−0.19, −0.04]). A label-free *closure gap*, motivated by the failure of mean-field closures that average a steep function over a heterogeneous neighborhood, is the share of the cohort's top 3% sites that are still in the top 3% after the fixed mean (the retention). It correlated with the effect of the fixed mean on the average precision over the 68 development and test cohorts (Spearman ρ = 0.43, p = 0.0002), but a choice between the fixed mean and no smoothing based on it, fitted on the development cohorts, selected the mean for 6% of the test cohorts and reached a mean AP of 0.36, against 0.37 for no smoothing (oracle choice 0.38) and 0.48 for similarity weighting, which makes

the choice unnecessary. As a robustness criterion in the spirit of grid convergence, the range of the average precision over the six anomaly shapes was 0.06 without smoothing, 0.22 for the fixed mean and 0.06 for the similarity-weighted rule.

4.6. GaaR against random search

Table 2 lists the best development configuration of every family. Among the 460 distinct configurations the frozen fixed mean ranked 455th on the development fitness (0.13; optimum 0.41). With 40 evaluations, random search found a configuration in the top 5% in 94% of the runs, the GaaR strategies in 70% (greedy), 81% (restart) and 80% (population), and in the top 1% in 67% against 53%, 58% and 59%: GaaR was not better than random search here, because many configurations of the similarity-weighted family are nearly equivalent. The champions of the restart strategy and of random search were the same configuration (similarity-weighted, window 3, persistence 0.5; $F = 0.31$), no better than the untuned rule. An earlier search of the scoring configuration itself (smoothing, axes, PCA dimension, rarity scale; 360 configurations; repository) gave a gain on fresh scenarios of +0.08 with a 95% interval of [−0.05, +0.17], which includes zero.

5. Limitations and future work

The lattices and the images are simulated and were written by the author of the method (the simulation is MyUncle's and its parameter sets are not fitted to measurements; the controlled generator has diverse, separable anomalies and Gaussian noise); nothing here shows that the method finds anything in an experimental system. Development and test data come from the same generator families, 16 to 50 independent units per test scenario are few, the rule parameters are selected with labels, and the handcrafted-feature images are not helped by any rule. Novelty is not a defect: a rare state may be a rare preparation or a known feature that is merely uncommon. The comparison of the rarity score itself with other detectors (isolation forest, local outlier factor, nearest-neighbor and Mahalanobis distance) was made under the earlier fixed smoothing and is in the repository: a smoothed Mahalanobis baseline scored higher than the rarity score on the development images, so the score is not claimed to be the best detector, and the contribution concerns the aggregation step. Planned: validation on a public data set and on experimental micrographs with independently established defects, blinded expert review, more units per scenario, other filters of the bilateral and non-local-means families as baselines, and a label-free choice between smoothing and no smoothing (the autocorrelation of the unsmoothed score is a candidate; in a first look it separated the two lattice scenarios in which the fixed mean destroyed the signal from the rest, with eight scenarios and one threshold, which is a hypothesis, not a rule).

6. Conclusions

A fixed smoothing window can destroy an anomaly signal, depending on the size, shape and scatter of the anomalies and on the embedding. Aggregation weighted by embedding similarity, with a label-free scale, was a robust replacement on independent test scenarios (paired intervals exclude zero). Gates based on local order and on the von Mises–Fisher information, the d-dimensional extension of MyUncle's entropy quantities, did not improve on ordinary similarity, and an evolutionary search over rule configurations did not beat random search. All evidence is simulated; experimental validation is the next step.

Declarations

Funding: no external funding. **Competing interests:** the author declares no competing interests. **CRediT:** Leon Sandler, sole author: conceptualization, methodology, software, validation, formal analysis, investigation, visualization, writing – original draft, writing – review and editing. **Data and code availability:** code, tests, benchmarks, search tables, seeds and results are open at `https://github.com/sandlerleon/MyUncleVision` (MIT license; vendored MyUncle and GaaR cores unmodified, GaaR commit 4021e64) and archived at `https://doi.org/10.5281/zenodo.24066260` (all versions). No experimental, patient or tissue data were used. **Generative AI:** during the preparation of this work the author used Claude (Anthropic) to assist with writing and testing the code, running the analyses and drafting the text and figures; the author reviewed and edited the content and takes full responsibility for it.

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Table 3: Independent test scenarios: average precision of aggregation rules. Columns: synthetic images with handcrafted features (Hand.) and DINOv2 (DINO); MyUncle lattices with the rotor (Rot.) and rotor + D (Rot.+D) embedding; controlled generator with scattered sites (Scat.), 3×3 blocks (Blk3), 6×6 blocks (Blk6), bars (Bar), irregular clusters (Irr.) and boundary blocks (Bnd.). The last column is the lower-tail fitness F with its 95% interval.

Rule	Hand.	DINO	Rot.	Rot.+D	Scat.	Blk3	Blk6	Bar	Irr.	Bnd.	F [95% interval]
No smoothing	0.18	0.11	0.56	0.43	0.37	0.34	0.38	0.39	0.39	0.34	0.23 [0.18, 0.31]
Fixed 3×3 mean (frozen)	0.04	0.50	0.32	0.46	0.03	0.10	0.25	0.05	0.08	0.08	0.11 [0.10, 0.14]
Similarity-weighted 3×3 (untuned)	0.17	0.75	0.51	0.47	0.47	0.44	0.47	0.50	0.48	0.45	0.32 [0.24, 0.43]
GaaR / random-search champion	0.17	0.64	0.52	0.47	0.47	0.43	0.46	0.49	0.47	0.44	0.31 [0.24, 0.42]
One-sided (dev-selected)	0.18	0.11	0.57	0.43	0.34	0.34	0.37	0.40	0.39	0.34	0.23 [0.18, 0.31]
Order gate, compatibility (dev)	0.18	0.44	0.57	0.48	0.45	0.41	0.43	0.47	0.45	0.42	0.30 [0.22, 0.40]
Order gate, R_i or H_χ (dev)	0.18	0.48	0.54	0.46	0.45	0.41	0.43	0.47	0.46	0.42	0.31 [0.22, 0.41]

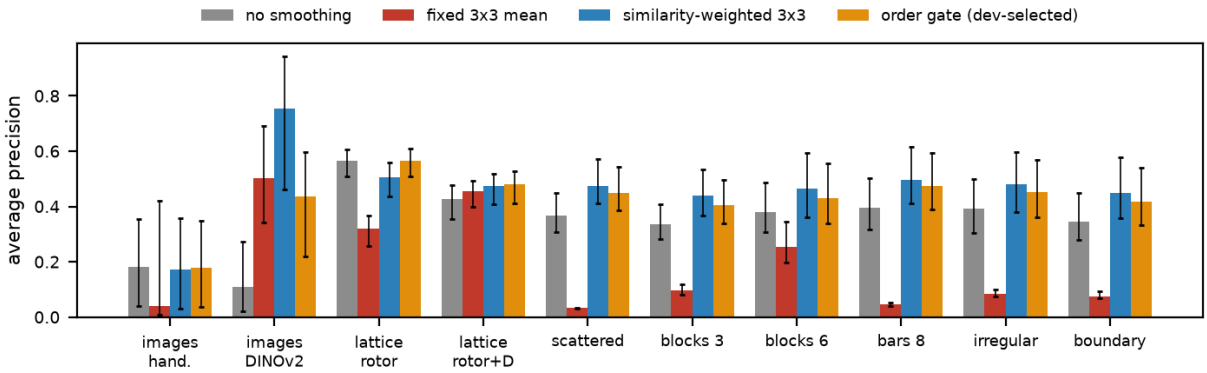


Figure 2: Average precision on the ten independent test scenarios with 95% intervals (resampling images or simulation runs).

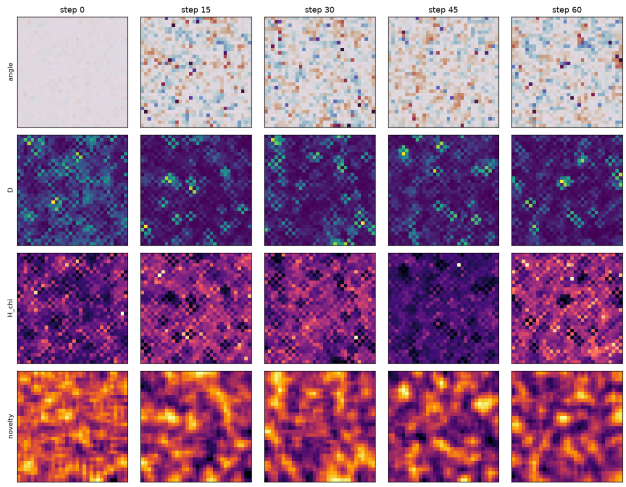


Figure 3: A MyUncle simulation analyzed with MyUncleVision (chiral scaffold preset, 32×32 sites). Rows: rotor angle, local disorder D, order information H_χ ($d = 2$) and the earlier frozen novelty score (rarity, fixed 3*imes*3 mean); columns: steps 0, 15, 30, 45 and 60.

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